REVISION 2021

DC Machines Lab

Official SITTTR Practical Handbook

Code: 3036

Credits: 1.5

Semester: 3

Type: Practical Core

Course Outcomes (CO)

- **CO1:** Identify the construction of DC machines and develop magnetic curves. (9 Hours)
- **CO2:** Develop the performance characteristics of various types of DC generators. (12 Hours)
- **CO3:** Apply various speed control techniques in dc motors to plot the speed curve and testing of dc machines. (9 Hours)
- **CO4:** Develop the performance characteristics of various types of DC motors. (9 Hours)

Curriculum Map

Module	Description	Hours	CO Mapping	Cognitive Level
1	Construction & OCC	9	CO1	Applying
2	Generator Characteristics	12	CO2	Applying
3	Speed Control & Testing	9	CO3	Applying
4	Motor Characteristics	9	CO4	Applying

Module 1: Construction & OCC

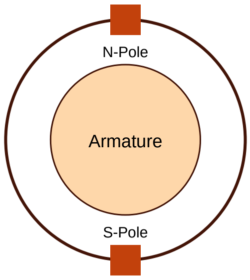
M1.01: Machine Assembly

Dismantle and reassemble a DC machine and identify parts like Commutator, Brushes, Armature, and Field poles.

M1.03: OCC Plotting

Run a separately excited and self-excited DC generator at rated conditions and plot Open Circuit Characteristics (OCC).

SVG Illustration: DC Machine Cross-section



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(OCC) .

Module 2: Generator Characteristics

M2.02: Shunt Generator Characteristics

Develop the internal and external characteristics of a DC shunt generator at constant speed.

M2.04: Compound Generator

Plot efficiency curves of DC compound generator in Long shunt and Short shunt configurations.

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Module 3: Speed Control & Testing

Speed Control Techniques

Armature control and Field control methods for DC shunt motors.

Swinburne's Test

Pre-determination of efficiency of a DC machine by Swinburne's test.

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Practice Model Question Paper

Practice Model Paper — not an official REV2021 paper

Part A: OCC

Perform the OCC test on a separately excited DC generator and find the critical resistance at 1500 RPM.

Part B: Speed Control

Plot the speed vs armature voltage characteristic of a DC shunt motor by keeping field current constant.

Source Declaration

Curriculum Scheme: Revision 2021

Subject Name: DC Machines Lab

Subject Code: 3036

Official Source URL: [SITTTR Link]

Source Verification Date: 2026-08-19

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